Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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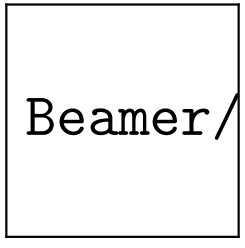
Título da Apresentação

Seu Nome

Universidade Federal de Mato Grosso
Faculdade de Engenharia
Engenharia de Computação

Orientador:
Prof. Dr. Fulano Beltrano da Silva

Trabalho de Conclusão de Curso
19 Pebruari 2025



Conteúdo

Introdução

Recursos do LaTeX

Conclusão

Introdução

Este *tamplate* é feito com a logo oficial da FAENG-UFMT. É uma versão modificada do *tamplate* da Universidade Politécnica da Catalunha (UPC). ▸ Link

Para aprender a construir apresentações em LaTeX, veja o seguinte tutorial: ▸ Link

Figuras



Tabelas

Modelo	Acurácia
SVM	0.88
k-NN	0.80
Naive Bayes	0.83
MLP	0.95

Exemplo de Tabela

Listagem

Listagem de itens com *itemize*:

- Item 1
- Item 2
- Item 3

Listagem de itens com *enumerate*:

1. Item 1
2. Item 2
3. Item 3

Blocos

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Notação Matemática

Distribuição Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Transformada de Fourier Discreta

$$X(\omega) = \sum_{n=0}^{L-1} x_n e^{-i\omega n}, \quad x \in \mathbb{R}^L$$

Algoritmos

Algoritmo *Selection Sort*, que ordena um vetor $x \in \mathbb{R}^N$ em tempo $\mathcal{O}(N^2)$:

$i \in [0, ..., N-1]$ $m \leftarrow i$ $j \in [i+1, ..., N-1]$ $x_j < x_m$ $m \leftarrow j$
$x_i \neq x_m$ $a \leftarrow x_i$ $x_i \leftarrow x_m$ $x_m \leftarrow a$

Código-fonte

Código-fonte do algoritmo *Selection Sort* em Python:

```
lista = [3,2,1]
for i in range(len(lista)):
    menor = i
    for j in range(i+1,len(lista)):
        if lista[j] < lista[menor]:
            menor = j
    if lista[i] != lista[menor]:
            aux = lista[i]
            lista[i] = lista[menor]
            lista[menor] = aux
print(lista)
```

Utilize "fragile" ao invés de "plain" ao definir um *frame* que contém um *verbatim*

Obrigado!